

Yongjun Rong

Cherry Hill, NJ, USA, 08003 (US Citizen) • Email: rongyj@hotmail.com • Tel: 856-408-4027
Linkedin: <https://www.linkedin.com/in/rongyongjun/>, Github: <https://github.com/rongyj>

EDUCATION

Jun., 2006, Master of Computer Science, Texas Tech University, Lubbock, TX, USA
Feb., 1996, Master of Engineering, Shanghai JiaoTong University, Shanghai, China
Jul., 1991, Bachelor of Engineering, Changsha Railway University, Changsha, Hunan, China

SUMMARY

Yongjun Rong is a hands-on visionary cloud architect and technology leader with deep expertise in GenAI systems, prompt engineering, cloud infrastructure, and digital transformation. With over two decades of experience spanning finance, consulting, SaaS, and technology sectors, he has architected and delivered large-scale distributed systems, led mainframe-to-cloud migrations, and pioneered automation and DevOps best practices. Yongjun excels at designing resilient, scalable microservices, integrating enterprise SaaS applications, and driving innovation through mentorship and cross-team enablement. His technical breadth covers AWS, GCP, Azure, Java, Python, Node.js, Kubernetes, Terraform, and more. Fluent in both English and Chinese, he is passionate about empowering organizations to achieve agility, reliability, scalability and business impact through modern cloud-native solutions.

SKILLS

- GenAI system architecture and implementation with Retrieval-Augmented Generation (RAG) and multi-agent orchestration to accelerate software development and mainframe migration to modern cloud-native platforms.
- Cloud infrastructure design, automation, and migration across AWS, GCP, and Azure, including serverless architectures (Lambda, EventBridge, Step Functions), container orchestration (EKS, AKS, GKS), and Infrastructure as Code (Terraform, Terragrunt, CloudFormation, Ansible, Puppet, Chef).
- Enterprise architecture leadership for large-scale finance, insurance, and telecommunications organizations, specializing in distributed systems, microservices, event-driven architectures, streaming data processing, and technology strategy development.
- Digital transformation engineering, refactoring legacy monolithic systems to cloud-native, loosely-coupled distributed architectures using Java, Python, Node.js, SpringBoot, and modern data stores (SQL, NoSQL, DynamoDB, MongoDB, ClickHouse, PostgreSQL, Oracle).
- DevOps and automation expertise in cloud and on-premises environments, including CI/CD pipeline design, configuration management, system/network administration, and orchestration with Docker, Kubernetes, Jenkins, Bamboo, and Linux HA technologies.
- Integration architecture and implementation for SaaS enterprise applications (ERP, CRM, Accounting, HR), leveraging VPC Endpoints, PrivateLink, REST/SOAP APIs, WSDL/XML/JSON, and secure authentication/authorization mechanisms.
- Design and delivery of highly available, scalable, and resilient microservices and event-driven systems, applying best practices in CQRS, event sourcing, domain-driven design, and cloud-native patterns for internet-scale applications.
- Mentorship and leadership in technology adoption, process automation, and cross-team enablement, fostering a culture of innovation, reliability, and continuous improvement across engineering organizations.

HOBBIES

- Pickleball (~4.0 player)
- Pingpong (ITTF Score around 1200)
- IoT or Robot programming (Have iRobot and MIPOSaur at home. Programming against iRobot create SDK and MIPOSaur SDK.)
- iOS/Android Apps Development (Developing a Chinese character learning App)
- Playing with cutting-edge technologies (genAI/Autogen/Multiagents, Docker, Go, Kafka, Kubernetes and Amazon CloudFormation, JavaScript, AngularJS, ReactiveJS, Node.js)
- IFTTT Home Automation (IFTTT, WeMo, EcoBee, Weather Underground, SaaS, REST, Home Automation)

WORK EXPERIENCE SUMMARY

Employments	Major Accomplishments & Technologies
Oct. 2025 - Current Intuitive.ai Cherry Hill, NJ (WFH), USA Principal Cloud and Platform Architect	- EKS Service Mesh (Istio) on-demand framework Design and Implementation - Proprietary Environment As Code (EAC/IAC) self-teaching, mentoring and troubleshooting <i>IaC, Terraform, AWS/EKS/Service Mesh/Istio, TLS Certificate</i>
Feb. 2025 - Oct. 2025 Biophy Cherry Hill, NJ (WFH), USA Director of Cloud Architecture	- Comprehensive Infrastructure as Code (IaC) Design and Implementation - AI-Powered Data Ingestion and Scraping System for RAG - Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS <i>IaC, Terraform/Terragrunt, AWS/EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager/Lambda/CloudWatch/EventBridge/SQS/SNS, Python/Django/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture</i>
Dec. 2021 - Feb. 2025 McKinsey & Company Cherry Hill, NJ (WFH), USA Senior Cloud Engineer II (Associate)	- Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks - Using genAI (openAI) with multi-agents to accelerate the migration of mainframe to modern technologies - Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc. <i>genAI/openAI/Langchain/RAG/Multi-agents/prompts tuning, VectorStore/ChromaDB/Milvus, Python/Flask/FastAPI/Pydantic/Jupyter, Databricks/pyspark, Typescript/JavaScript/NodeJs, Event Driven Architecture (EDA), Terraform, Microservices, Java/SpringBoot/NodeJs/Python/Flask/FastAPI, EventBridge/CloudWatch/Lambda/Step Functions, DynamoDB/RDS/Oracle, Docker/K8s/EKS/AKS, Json Schema, JOI, OpenAPI/Swagger</i>
Dec. 2020 - Dec. 2021 Yotascale Cherry Hill, NJ (WFH), USA Staff Engineer	Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform <i>OpenAPI/Swagger, Microservice, Java/Kotlin/SpringBoot, Kubernetes/EKS/AKS, Istio, Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, Python/Bash/Go, ELK/AppDynamic/Datadog</i>
Mar. 2020 - Nov. 2021 Gigster Inc. Cherry Hill, NJ, USA Technical Infrastructure/Devops Architect	Images manipulation/detection software evaluation system <i>Microservice, NodeJs, Kubernetes/EKS, Terraform, RabbitMQ/ActiveMQ, Docker, AWS(EC2/IAM/S3/EKS/ALB/RDS/VPC/R53/AutoScaling/SES/ECR..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins, JavaScript</i>
Jun. 2017 - Feb. 2020 Barclays Wilmington, DE, USA VP,Lead Architect Engineer	- Delivery Lead for strategic distributed system technology domain - Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud <i>Terraform, CQRS, Event Store, Event Driven Architecture (EDA), Kafka/ActiveMQ/IBM MQ, Kafka Stream, Docker, AWS(EC2/IAM/S3/ECS/EKS/ELB/Lambda/RDS/VPC/R53/Secret Manager/AutoScaling/SES..etc), Puppet/Chef/Ansible, Git/Maven/Jenkins/Bamboo, Python/Ruby/Scala/Bash/Go, ELK/AppDynamic, DynamoDB/MongoDB/ElasticCache/GridGain, Linux/Windows/AIX/Solaris, aPaaS/VMWare/Citrix</i>
May 2014 - June 2017 Oracle Redwood, CA (WFH@NJ), USA Lead Software Engineer	- Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors - Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite - Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework <i>Java, SOA, Web Services, SOAP/WSDL, REST/JSON/XML, XSchema/XSD, DOM, WSDL4J, SDK, API, MVC, Docker, Cloud, Amazon EC2, NoSQL, GIT, Maven, Junit, SoapUI, IaaS, SaaS, AWS</i>

Employments	Major Accomplishments & Technologies
Oct. 2010 - Apr. 2014 Comcast T&P Philadelphia, PA, USA Senior Cloud DevOps/Infrastructure Automation Engineer	• AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven • Configuration management and deployment automation with Puppet and Capistrano • Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins • Performance Monitoring System for Jetty, JVM, Linux and Splunk • Secured FTP server with virtual users and admins users with complex access permissions • BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance • Two Steps bare-metal Linux bootstrap system for HP Servers • Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers • Automation scripts and tools for DevOps • CoreOS and Docker prototyping and testing • SDLC process management and junior team member training and mentoring <i>DevOps, Automation, Docker, CoreOS, Ruby, Ruby on Rails 3, Bash, SDLC, GIT, Maven, Jira/Confluence, Bamboo/Jekins/Hudson, Nexus/Artifactory, Gerrit, CI/CD, Performance Monitoring, Splunk, JMX, Capistrano, PXE/TFTP, DHCP, DNS, Cobbler, Centos, MySQL/PostgreSQL, NoSQL, Cassandra, RabbitMQ, JMS, LDAP, SSO, Metrics, Graphite, Agile, RPC, Cloud, Clustering, Linux HA, HAProxy, Pacemaker/Corosync/Cman, SAN, GFS2, RAC, Volume Manager</i>
Nov. 2006 - Oct. 2010 Boomi Inc. Berwyn, PA, USA Senior Software Engineer	• Boomi On-Demand Business Integration Cloud Service • Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment • Desktop Accounting and ERP Enterprise Applications Integration (EAI) • Amazon EC2 and S3 SOA HA Environment • Setup Development, Build and Deployment Environment for QA, Staging and Production Environment • Research virtualization technology and implement Virtual Machines using VMware server and workstation • Java to C/C++, DLL, COM/COM+/DCOM/.NET Integration • SaaS system integration • SAP system integration <i>Java, SOA, IaaS, PaaS, EAI, XML, XSD, JAXB, JavaScript, .NET, EC2, S3, EBS, REST, SOAP, WSDL, SaaS, SAP, BAPI, ABAP, MySQL/PostgreSQL, Maven, SVN, JVM, JVM GC Tuning, Apache Lucene, Apache Solr, VMWare, Cloud</i>
Jun. 2006 - Nov. 2006 Vantage learning Newtown, PA, USA Senior System Engineer Manager	• Large scale SOA e-learning system architecture design and blueprinting • Large Data Center Design and Administration • Sun SPARCStorage SSA Array 100 and Sybase SQL Server (10.x, 11.x) Data (DB schema, data and Stored-Procedure) Recovery • Ecommerce Open Source Software installation and Configuration • Resin Application Server Distributed Session Design and Setup <i>Linux, Centos, SOA, Java, .NET, Network, VLAN, Router, Firewall, Load Balancer, SOAP/WSDL, MySQL/PostgreSQL, Struts, Disk Array, Fedora, Redhat Linux, LAMP, eCommerce, SQL</i>

Employments	Major Accomplishments & Technologies
May 2002 - Jun. 2006 Computer Science, Texas Tech University Lubbock, Texas, USA Senior Programmer/Analyst and Unix System Administrator	• Unix network System and Software routine maintenance • Setup Veritas Netbackup System with RAID0/1/5 and Tape Library • Setup LDAP, Kerberos KDC and Cross-Realm system • Migrated email system from UNIX mail server to Windows Exchange Server • Secured Postfix SMTP server with SASL Authentication and Verification • Installed AIX 5.1 and configured NIM on IBM RS6000 (10x 7043-140 boxes and 1x 7024-E30 box) • Deployed Globus Toolkit3.2.1 (OGSA/OGSI) Grid Computing Environment on Solaris 9(3 boxes) and Linux (3 boxes) • Setup Jumpstart Environment for Solaris 9 and Solaris 10 • Designed and Implemented the Single Sign-On (SSO) for Unix and Windows Environments • The Migration from NIS+ to LDAP for Solaris <i>JAVA, C/C++, GCC, GNU Make, perl, LDAP, NFS, SSO, Kerberos, SSL, SMTP, Email Server and Client, Solaris, AIX, Mac OSX, Linux, Windows Server, PAM, RAID, SCSI, Veritas, TFTP, RARP/ARP, Bootparamd, DHCP, Postfix, PostgreSQL, NIS/NIS+</i>
Aug. 2002 - Jun. 2006 Computer Science, Texas Tech University Lubbock, Texas, USA Master Student	• Master Thesis: A Service Wrapping and Provisioning Framework for SOA • GIS Project: A Yield Mapping and Prediction Information Delivery System • Independently finished practical course projects list <i>Java, C/C++, GCC, Gnu Make, Linux, Linux Kernel, GIS, Ant, Struts, J2EE, UML2, MVC, Maven, PostGIS, PostgreSQL, Solaris</i>
Apr. 2000 - May 2002 Technology Development Center, Sun Microsystem (China) Co.,Ltd. Beijing, P. R. China Senior Software Engineer	• Java Message Service(JMS) Prototype Application • Remote Learning Management Platform (www.ambow.com) • Bao Steel B2B Website (www.bsteel.com) • Project Management Tools <i>Java, J2EE, JMS, EJB, Servlets, JDBC, Message-Driven Bean, Oracle DBMS, Java IDL, RMI, CORBA, JNDI, Solaris, CVS, Ant, LDAP, SQL, Weblogic, OOA/OOD, UML, MVC</i>
Aug. 1998 - Apr. 2000 IBM China Research and Development Lab Beijing, P. R. China Software Engineer	• Credit Card Internet Online Payment System • Jiro/Jini-based Storage Management System • Handwriting Recognizing System <i>Java, J2EE, C/C++, Servlets, WebSphere, SSL, JDBC, Windows NT, SAN, POS, JNI, DB2</i>
Apr. 1996 - Jul. 1997 North China Institute (Taiji Computers Corporation) Beijing, P. R. China System/Network/Software Engineer	• System Engineer, IBM Service Center • Software Engineer, HUAXU Golden Card Co.Ltd. • Large scale Network System Design and Administration <i>C/C++, Assembly, Network Design, Router, Switch, VLAN, AIX, Disk Array, SSA, RAID, Volume Manager, IC, RIP/OSPF, Ciso IOS</i>
Jun. 1991 - Sep. 1993 Yongji Electrical and Motor Factory ShanXi Province, P. R. China Mechanical Engineer	• CIMS and FMS CAPP/CAD/CAM manufacturing Automation application development <i>C/C++, AutoCAD, CAD/CAM/CAPP, CIMS/FMS, DOS</i>

SELECTED PROJECTS

1. Oct. 2025 - Current, Principal Cloud and Platform Architect, Intuitive.ai, Cherry Hill, NJ (WFH), USA

As a hands-on of principal cloud and platform architect, I designed and implemented eks, service mesh framework to support the development of on-demand platform for hundreds of developers. Proprietary environment as code (EAC) self-teaching and mentoring teammates.

1.1 EKS Service Mesh (Istio) on-demand framework Design and Implementation

Architected and implemented a comprehensive EKS-based service mesh framework using Istio to enable on-demand platform capabilities for hundreds of developers. Designed a scalable, secure, and automated infrastructure that supports dynamic service provisioning, traffic management, and observability across microservices.

Highlights:

- Designed and implemented EKS cluster architecture with Istio service mesh for multi-tenant on-demand platform
- Developed automated framework for dynamic service mesh provisioning and configuration management
- Implemented mTLS (mutual TLS) for secure service-to-service communication across all microservices
- Configured Istio VirtualServices and DestinationRules for advanced traffic routing and load balancing
- Designed and implemented centralized certificate management system for TLS/mTLS certificates
- Built comprehensive observability stack with distributed tracing, metrics, and logging integration
- Created Terraform modules for automated EKS and Istio deployment and configuration
- Implemented service mesh policies for security, rate limiting, and circuit breaking
- Designed multi-namespace isolation strategy for secure multi-tenant operations
- Established CI/CD pipelines for automated service mesh configuration updates and deployments

Technologies: AWS EKS, Istio Service Mesh, Kubernetes, Terraform, TLS/mTLS, Envoy Proxy, Prometheus, Grafana, Jaeger, Kiali, Helm, AWS IAM/RBAC

Leadership:

- Led the architecture and implementation of the service mesh framework that supports hundreds of developers on the on-demand platform.
- Established service mesh best practices and patterns that improved security, observability, and operational efficiency.
- Mentored platform and development teams on Istio service mesh concepts, traffic management, and security policies.

1.2 Proprietary Environment As Code (EAC/IAC) self-teaching, mentoring and troubleshooting

Self-taught proprietary infrastructure as code software named Environment As Code (EAC) and mentoring team members and troubleshooting complex AWS infrastructure configuration issues.

Highlights:

- Designed and implemented EKS cluster architecture with Istio service mesh for multi-tenant on-demand platform
- Developed automated framework for dynamic service mesh provisioning and configuration management
- Implemented mTLS (mutual TLS) for secure service-to-service communication across all microservices
- Configured Istio VirtualServices and DestinationRules for advanced traffic routing and load balancing
- Designed and implemented centralized certificate management system for TLS/mTLS certificates
- Built comprehensive observability stack with distributed tracing, metrics, and logging integration
- Created Terraform modules for automated EKS and Istio deployment and configuration
- Implemented service mesh policies for security, rate limiting, and circuit breaking
- Designed multi-namespace isolation strategy for secure multi-tenant operations
- Established CI/CD pipelines for automated service mesh configuration updates and deployments

Technologies: AWS EKS, Istio Service Mesh, Kubernetes, Terraform, TLS/mTLS, Envoy Proxy, Prometheus, Grafana, Jaeger, Kiali, Helm, AWS IAM/RBAC

Leadership:

- Led the architecture and implementation of the service mesh framework that supports hundreds of developers on the on-demand platform.
- Established service mesh best practices and patterns that improved security, observability, and operational efficiency.
- Mentored platform and development teams on Istio service mesh concepts, traffic management, and security policies.

2. Feb. 2025 - Oct. 2025, Director of Cloud Architecture, Biophy, Cherry Hill, NJ (WFH), USA

As a hands-on director of cloud architecture, I designed and implemented comprehensive infrastructure as code (IaC) by importing existing AWS resources and redesigning them according to AWS best practices. I led the development of a data ingestion and scraping system for an AI-powered RAG platform, following SOLID principles and Clean Architecture patterns. Additionally, I created a custom AWS CloudWatch custom alerting system leveraging Lambda, SQS and SNS for robust monitoring and notification.

2.1 Comprehensive Infrastructure as Code (IaC) Design and Implementation

Led the transformation of AWS infrastructure to infrastructure as code (IaC), importing and redesigning resources to align with best practices. Developed modular Terraform configurations and established secure, standardized cloud operations.

Highlights:

- Imported and catalogued existing AWS resources including EC2 instances, VPC configurations, security groups, and IAM policies
- Redesigned infrastructure architecture following AWS Well-Architected Framework principles

- Implemented modular Terraform configurations with reusable modules for consistent deployment
- Established comprehensive security controls including least-privilege IAM policies and network segmentation
- Created automated validation and testing procedures for infrastructure changes
- Documented infrastructure patterns and best practices for team adoption

Technologies: Terraform, AWS EC2/EBS/S3/VPC/ALB/Security/R53/IAM/Secrets Manager, Infrastructure as Code, AWS Best Practices, Security Controls

Leadership:

- Led the transformation from manual infrastructure management to fully automated IaC practices.
- Established infrastructure standards and patterns that improved team productivity and security posture.
- Mentored team members on AWS best practices and Terraform/Terragrunt development.

2.2 AI-Powered Data Ingestion and Scraping System for RAG

Architected and delivered a robust data ingestion and scraping platform for AI-powered RAG, applying SOLID and Clean Architecture principles to ensure scalability, maintainability, and high data quality.

Highlights:

- Architected a modular data ingestion pipeline supporting multiple data sources and formats
- Implemented intelligent web scraping capabilities with rate limiting and error handling
- Designed data transformation layers following Clean Architecture principles
- Built robust error handling and retry mechanisms for data processing failures
- Implemented comprehensive logging and monitoring for data pipeline health
- Created automated testing frameworks for data quality and pipeline reliability

Technologies: Python/Django/Pydantic, openAI API/LangChain, SOLID Principles, Clean Architecture, Data Processing, Web Scraping

Leadership:

- Drove the adoption of Clean Architecture and SOLID principles across the data engineering team.
- Established data quality standards and monitoring practices for AI system reliability.
- Mentored developers on modern software design patterns and best practices.

2.3 Custom AWS CloudWatch Alerting System with EventBridge, Lambda, SQS and SNS

Developed a proactive monitoring and alerting system using AWS CloudWatch, Lambda, SQS, and SNS, enabling automated detection and response to critical events across cloud infrastructure.

Highlights:

- Designed custom CloudWatch metrics and dashboards for comprehensive system visibility
- Implemented SQS-based message queuing for reliable alert processing and delivery
- Configured SNS topics and subscriptions for multi-channel notifications (email, SMS, Slack)
- Built automated response workflows triggered by specific alert conditions
- Created alert escalation policies based on severity and time-based thresholds
- Implemented alert correlation and deduplication to reduce alert fatigue

Technologies: AWS CloudWatch, SQS/SNS, Lambda, EventBridge, Monitoring, Alerting

Leadership:

- Established comprehensive monitoring and alerting standards across the organization.
- Reduced mean time to detection (MTTD) and mean time to resolution (MTTR) through proactive monitoring.
- Trained operations teams on monitoring best practices and incident response procedures.

3. Dec. 2021 - Feb. 2025, Senior Cloud Engineer II (Associate), McKinsey & Company, Cherry Hill, NJ (WFH), USA

Led the development of genAI systems with RAG and multi-agents to accelerate mainframe migration to modern technologies. Built comprehensive genAI solutions with RAG to speed up data pipeline development on PySpark and Databricks for complex data transformations. Scaled RAG vectorstore (ChromaDB) for a call center chatbot copilot. Successfully migrated and refactored complex EDA mainframe systems to AWS serverless architecture using EventBridge, Lambda, Step Functions, SQS, SNS, CloudWatch, and EKS.

3.1 Using genAI (openAI) with RAG (ChromaDB, Milvus) to accelerate the development of data pipelines running on pyspark and databricks

Designed and implemented a genAI system with RAG (ChromaDB/Milvus) to automate and accelerate the development of data pipelines, enabling seamless transformation of source tables to target tables with varying data structures.

Highlights:

- Prompts tuning to generate data pipeline pyspark function code and unittest code
- Design and implement chunking system (by target table or by relationship) to split the large inputs into small chunks to avoid overflowing the openAI API token limit
- Design and implement parallel calls to openAI API by using asyncio with yield to improve performance
- Implement FastAPI API to expose source to target data transformation as REST service
- Design and document the local development and debug workflow by using debugpy with VSC and pycharm

Technologies: Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Python/Pydantic/Asyncio/Flask/FastAPI/Debugpy, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker/Compose, PostgreSQL

Leadership:

- Spearheaded the integration of genAI and RAG for data pipeline automation, enabling rapid transformation of complex data structures.
- Mentored engineers on advanced prompt engineering and parallel AI workflow design, raising team proficiency in cutting-edge AI techniques.
- Established robust local development and debugging standards, improving team productivity and onboarding efficiency.

3.2 Using genAI (openAI) with multi-agents to accelerate the migration of mainframe to modern technologies

Built a multi-agent genAI platform to automate and accelerate the migration of mainframe applications to modern technologies, integrating RAG for context-aware transformation and process automation.

Highlights:

- Core Contributors to genAI system to migrate mainframe frontend technologies (BMS/JSF/JSP) to modern technologies (React/Angular..etc.)
- Contribute to develop multi-agents genAI system to automate the development/testing/deployment process
- Contribute to develop RAG system to accelerate the migration process from Cobol to Java by using ChromaDB/Milvus to store the dependency context

Technologies: Langchain/VectoreStore/ChromaDB/Milvus, genAI/openAI, RAG, Agents/Multi-agents/Autogen, Python/FastAPI, Typescript/JavaScript/NodeJs, Microservice, Dockerfile/Docker Compose, PostgreSQL

Leadership:

- Drove the integration of multi-agent genAI systems to automate legacy migration, reducing manual effort and error rates.
- Collaborated across frontend and backend teams to ensure seamless modernization from mainframe to cloud-native stacks.
- Championed the use of RAG for context-aware migration, setting a new benchmark for future modernization projects.

3.3 Migration/Refactoring of Mainframe (IBM MQ) EDA system to AWS by using EventBridge/Lambda/Step Function/CloudWatch/EKS/SQS/SNS...etc.

Engineered and deployed an EKS-based router to integrate AWS EventBridge with external MQ events, providing robust monitoring, reporting, and cloud-native migration for mainframe EDA systems.

Highlights:

- Develop SpringBoot EKS router to convert XML/XSD to JSON schema and integrate with MQ events
- Design and implement the monitoring terraform module for all different AWS resources (Lambda, EventBridge, S3, DynamoDB, Step Function..etc.) using cloudwatch metrics
- Design and implement the reusable report generator using Lambda and EventBridge and deployed as Microservices which can be called to run any custom AWS CloudWatch insights queries in custom schedule

Technologies: Java/SpringBoot, Terraform, Microservice, Java/JavaScript, EKS, AWS/EventBridge/CloudWatch/Lambda, XML/XSD/Json/Json Schema, DynamoDB, RDS/Oracle

Leadership:

- Led the end-to-end migration of mission-critical EDA systems to AWS, ensuring zero downtime and robust cloud integration.
- Developed reusable infrastructure modules, empowering other teams to accelerate their own cloud migration journeys.
- Instituted comprehensive monitoring and reporting standards, elevating operational visibility and reliability.

4. Dec. 2020 - Dec. 2021, Staff Engineer, Yotascale, Cherry Hill, NJ (WFH), USA

Led the architecture and development of a comprehensive cloud cost analytics, optimization, and management platform supporting AWS, Azure, and GCP. Designed and implemented cross-cloud cost visibility, allocation, and optimization capabilities to help organizations manage cloud spending effectively.

4.1 Lead Architect and Lead Implementation Engineer for Cloud(AWS/Azure/GCP) Cost Analytics/Optimization/Management platform

Architected and implemented a cross-cloud cost analytics and optimization platform, enabling organizations to gain actionable insights and manage cloud spending across AWS, Azure, and GCP. Led the design of scalable data pipelines, analytics microservices, and secure infrastructure.

Highlights:

- Introduce and Build Microservice Design First Methodology and team culture based on OpenAPI/Swagger Specification
- Design and develop Cost Analytics/Lens Navigation/Lens Management Microservice using SpringBoot with ClickHouse and PostgreSQL as DW and DBMS
- Design and implement dynamic metadata driven widget/filter configuration Microservice
- Implement bitbucket CI/CD pipeline to build, deploy Microservices/GraphQL to eks/k8s with Istio
- Design the AWS infrastructure architecture with EKS/ALB/Istio/S3/SES/R53/ECR/PostgreSQL/RabbitMQ/SFTP
- Design/Implement terraform templates/scripts to deploy the full AWS infrastructure stack with IAM roles/users
- Develop terraform templates to deploy ALB controller/RabbitMQ and PostgreSQL by using helm provider
- Develop terraform templates to deploy k8s' CRDs, Pods, Services, Service Account by using Kubernetes-alpha and Kubernetes providers
- Design/Develop multiple terraform modules with Terragrunt.hcl to maintain the inputs/outputs dependencies between modules and keep configurations DRY
- Develop terraform templates to dynamically add R53 zone and records with TLS certificates validations
- Develop terraform templates to dynamically create S3 buckets with IAM policies to allow k8's service account to assume IAM role to access control using OIDC
- Develop terraform templates to create SFTP server backed with S3 bucket with IAM role and policy for sftp users
- Compile Microservice design/development guide lines based on OpenAPI/Swagger Spec
- Implement Junit test cases with local docker test containers
- Develop common multiple DBMS (ClickHouse and PostgreSQL) access lib with AWS secrets manager
- Implement json schema validation using open source

Technologies: Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ, Azure/GCP, Junit/Test Containers, Docker/Minikube

Leadership:

- Architected a cross-cloud cost analytics platform, aligning technical vision with business goals for multiple stakeholders.
- Introduced and enforced microservice design-first culture, elevating team standards and delivery quality.
- Mentored engineers in infrastructure-as-code and CI/CD best practices, fostering a culture of automation and reliability.

5. Mar. 2020 - Nov. 2021, Technical Infrastructure/Devops Architect, Gigster Inc., Cherry Hill, NJ, USA

Designed and implemented comprehensive AWS infrastructure architecture including CI/CD automation, Terraform infrastructure as code, and EKS container orchestration. Led the development of scalable, automated cloud infrastructure solutions to support microservices and distributed systems.

5.1 Images manipulation/detection software evaluation system

Led the architecture and implementation of an image manipulation and detection evaluation platform using microservices and event-driven architecture on AWS EKS. Automated infrastructure provisioning and deployment pipelines for scalable, efficient operations.

Highlights:

- Design and implement AWS infrastructure for Microservices (NodeJs) and EDA within EKS cluster
- Design and implement fully automated AWS stack in Iaas git repo by using terraform/terragrunt/bash
- Develop terraform scripts/modules to dynamically deploy/configure multiple Microservices with pods/containers/services to EKS cluster by using kubernetes terraform provider
- Develop terraform scripts to deploy/configure AWS ALB as eks ingress controller by using helm terraform provider
- Develop terraform scripts to customize IAM role/policy, k8s SA(service account) and CRD (Custom Resources Descriptor) for ALB Ingress Controller by using Kubernetes-alpha terraform provider
- Develop terraform templates to deploy RabbitMQ and PostgreSQL by using terraform helm provider
- Design/Develop multiple terraform modules with Terragrunt.hcl to maintain the inputs/outputs dependencies between modules and keep configurations DRY
- Develop terraform templates to dynamically add R53 zone and records with TLS certificates validations
- Develop terraform templates to dynamically create S3 buckets with IAM policies to allow k8's service account to assume IAM role to access control using OIDC
- Develop terraform templates to create SFTP server backed with S3 bucket with IAM role and policy for sftp users

- Develop terraform templates to dynamically create IAM user with customized group and policies

Technologies: Cloud Cost Analytics/Optimization/Management, OpenAPI/Swagger, Microservice, Java/Kotlin/Python, Spring/SpringBoot/Flask, Kubernetes/EKS/AKS/GKS, Terraform/Terragrunt/Ansible, AWS/EC2/EBS/S3/VPC/Security/R53/IAM/Secrets Manager/Lambda, Service Mesh/Istio/Envoy, Helm/Chart, RabbitMQ

Leadership:

- Designed and implemented a fully automated cloud infrastructure, setting a new standard for deployment efficiency.
- Enabled rapid scaling and reproducibility by modularizing infrastructure as code, empowering future project teams.
- Provided hands-on mentorship in DevOps automation, upskilling team members in Terraform and Kubernetes.

6. Jun. 2017 - Feb. 2020, VP,Lead Architect Engineer, Barclays, Wilmington, DE, USA

Directed the cloud adoption journey, migrating on-premises enterprise systems to AWS. Served as delivery lead for distributed system technology, including microservices, event-driven architecture, message transport, integration (batch/ETL), and event stream processing.

6.1 Delivery Lead for strategic distributed system technology domain

Directed the architecture and technology strategy for distributed systems, microservices, event-driven architecture, and stream processing, driving innovation and technology transformation across the organization.

Highlights:

- End to End Architecture; guiding solution architecture; developing design patterns Architecture Reviews - identifying reuse opportunities, manage technical debt Monitoring emerging technologies, industry trends and their applicability to the enterprise Capability roadmap and heatmap Drive/Foster Innovation Influencing Strategic Planning
- Drive technology transformation from on-premises to public cloud, multi-tiers monolithic to cloud-native Microservice system, strong ACID to BASE data consistency, Sync/Request/RPC driven to Async/Event driven architecture, tight coupling to loose coupling, ETL/Batch/Time to realtime streaming processing, vendor locked to resilient commodity system, vendor favor to OSS, vertical scaling to horizontal scaling
- Establish group-wide distributed system architecture and technology strategy
- Build PoC and Reference Architecture for common use cases for distributed system
- Evaluate around 200 existing products/technologies and decide the invest/divest house positions
- Create the architecture handbook for Microservice as group-wide guidance for solution Architect
- Contribute to and review the architecture handbook for EDA, Stream Processing and Integration
- Design vendor and technology selection and design principles/standards and best practices for the domain
- Review and approve the new request for product/technology in the domain
- Maintain the scope document and product list for the domain
- Develop the roadmap to adopt new products/technology and decommission legacy products/technology
- Lead working group (~30 members) forum to discuss and make decision on technology strategies

Technologies: Technology strategic Design and Stewardship, Distributed system, Domain Driven Design, Microservice, ACID/BASE, (Event) Stream /Batch Processing, Event Driven Architecture (EDA), Message Transports (Kafka, ActiveMQ, IBM MQ), Data Integration (ETL/ELT), Compensating transactions/Sagas, JMS/AMQP, Resilience and Performance, CQRS/SOA, CAP

Leadership:

- Directed a cross-functional working group to define and execute the distributed systems technology strategy.
- Authored the architecture handbook for microservices, influencing solution design across the organization.
- Drove technology transformation initiatives, guiding teams through cloud adoption and event-driven architecture.

6.2 Group-wide Cloud Adoption for Contact Center, Merchant Services and Fraud

Led cloud assessment and migration strategy for hundreds of on-premises applications, providing technical guidance and troubleshooting to ensure successful adoption of AWS cloud architecture.

Highlights:

- Build python automation tools to access AWS secret manager via boto3
- Using python boto3 to build tools to dynamically manipulate the IAM policies, roles and users
- Using terraform to dynamically deploy whole AWS infrastructure including EKS, IAM, DNS, ALB, S3 ...etc.
- Building and maintaining the EKS cluster via terraform and k8s yaml
- Defined and designed cloud strategy and roadmap/journey for Merchant services, Fraud and Contact Center
- Ensured ~500 on-premises application cloud migration journey
- Assess ~500 on-premises applications to determine the cloud migration readiness
- Provide guidance to application team to design cloud migration architecture

- Guide the .NET system to AWS migration
- Consulting to migration on-premises system to AWS ECS
- Collect system data from Abacus, ServiceFirst query and repo tools

Technologies: AWS, EC2, RDS, S3, ECS, VPC, ACL/Security Group/Routing table, ELB/NLB/ALB, Private Link, EBS

Leadership:

- Led the assessment and migration of 500+ on-premises applications to AWS, ensuring business continuity.
- Developed cloud migration roadmaps and provided hands-on guidance to application teams.
- Championed automation of cloud infrastructure, reducing manual errors and accelerating delivery.

7. May 2014 - June 2017, Lead Software Engineer, Oracle, Redwood, CA (WFH@NJ), USA

The Cloud Connectivity software engineering team is responsible to develop the Cloud Integration Connectivity SDK and Java Connectors which enable the SOA Suite and Oracle Integration Cloud Service (ICS) connectivity to any Oracle internal or external SaaS or On-Premises Enterprise Applications.

7.1 Docker Swarm/Compose/Stack CD/CI Test Environment for Cloud SDK/APIs/Connectors

Pioneered the use of Docker Swarm, Compose, and Stack to create scalable, automated CI/CD test environments for cloud SDKs, APIs, and Java connectors, streamlining development and testing workflows.

Highlights:

- Designed and setup complex Docker containers network with Overlay and MACVLAN drivers
- Created Dockerfile to build WebLogic docker images with different external modules and libraries
- Built Docker Hosts in Oracle Cloud VMs, Amazon EC2 and local VirtualBox by using Docker Machine
- Designed and built the pool of Docker Cluster Environment by using Docker Swarm
- Created Docker Stack file and deploy the service stack for multiple Microservice
- Developed docker-compose.yml files with extending services for the full service stack including dependency Web Services, DBMS, NoSQL and JMS
- Researched and prototyped the Docker Swarm and Docker Compose integration

Technologies: Docker, Docker Machine, Docker Compose, Docker Swarm, Docker Stack, Docker CLI (Rest APIs), WebLogic, Oracle Cloud VMs, AWS, JMS, NoSQL (Cassandra), DBMS (MySQL and PostgreSQL)

Leadership:

- Pioneered the adoption of containerization for CI/CD environments, reducing setup time and increasing test reliability.
- Developed reusable Docker orchestration templates, enabling rapid prototyping for SDK and connector teams.
- Trained team members on advanced Docker networking and orchestration concepts.

7.2 Cloud Enterprise Applications Integration(EAI) connectivity SDK/API plugin framework for Oracle SOA Suite

Developed a robust SDK and plugin framework for seamless integration between Oracle SOA Suite and a wide range of SaaS and on-premises enterprise applications, enhancing connectivity and extensibility.

Highlights:

- Designed and implemented the WSDL regeneration framework by rebuilding the internal data model (DOM) from the persisted integration flow artifacts files (JCA, WSDL and XSD) and the remote custom data
- Designed and implemented new APIs for the SDK to provide cache and pagination abilities for the backend persistent layer
- Designed and implemented NLS and localization framework for displayable strings within the SDK
- Troubleshoot the WSDL/XSD generation issues related to complex mixing namespaces and prefixes
- Develop utility methods to handle the namespaces and prefixes conflicts while combining the importing external schema to a single WSDL file
- Develop recursive logic to follow the XML element types to embed the undefined data structure if it has the same target namespace or create new schema if it has different target namespace
- Dynamically and Programmatically generate or regenerate the simplified integration WSDL from large set of source WSDL, schema XSD files and fresh custom data by using XDK/JAXP/WSDL4J/DOM
- Programmatically manipulate the single WSDL file generation from the source WSDL and schema XSD files with external schema file importing
- Programmatically handle the anonymous Complex and Simple XSD type for the integration WSDL generation and regeneration
- Design and test Oracle RCU DB descriptive XML files for new DB tables/sequences with primary and foreign keys
- Worked together with different connector development teams to joint debug the WSDL regeneration related issues

- Train and transfer knowledge for the WSDL regeneration framework to geographically distributed connector development teams

Technologies: Java, SOA, SOAP/REST, WSDL/JSON, XML, XML Schema, API, SDK, IaaS, RCU, JCA, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, NLS/L10, Oracle DB, Jira, Agile, ADE, GIT, Maven, Junit, XMLUnit, SoapUI, OSC, ERP, HCM, RightNow

7.3 Oracle Fusion Applications (FA) Business Event/Message-Driven Connector Framework

Engineered a business event and message-driven connector framework for Oracle Fusion Applications, enabling real-time event discovery, filtering, and integration with external systems.

Highlights:

- Design and implement the business event connector for FA (OSC/ERP/HCM)
- Build the design time web UI using the Oracle Web UI SDK which is building on the top of ADF
- Design and Implement the Business Event Connector Runtime
- Programmatically manipulate the payload to align with the changed schema in the integration WSDL
- Design and implement the custom event based on the generic event concepts
- Programmatically replace the xsi:anyType generic object type with the concrete object type selected by the end user from the web UI

Technologies: Java, FA, OSC/ERP/HCM, WSDL/SOAP, REST, JSON, XSI, XML Schema, XDK, JAXP, WSDL4J, DOM, WS-Security, Jersey REST/JSON, MVC, ADF, ADE, Maven, Junit, XMLUnit, SoapUI

Leadership:

- Architected a business event connector framework, enabling real-time integration for enterprise applications.
- Drove the adoption of event-driven patterns, empowering teams to build scalable, asynchronous integrations.
- Mentored developers on advanced payload manipulation and schema evolution.

8. Oct. 2010 - Apr. 2014, Senior Cloud DevOps/Infrastructure Automation Engineer, Comcast T&P, Philadelphia, PA, USA

As a senior engineer in BITT(Build Integration Test Team), provided DevOps support for multiple development teams and supporting the developers for SDLC infrastructure system design/implementation, and Build/Release/Deployment/SCM/CI automation. Designed, developed and implemented the AutoBuild system, OpenSource performance monitoring system for the SDLC systems, two steps bare-metal Linux bootstrap system for HP servers, Linux HA Clusters for SCM/Maven/CI/Crowd servers, Ruby/Rails software system maintenance and improvement.

8.1 AutoBuild System for Build/Release more than 200 Microservice projects with GIT/SVN/Maven

Technologies: Ruby/Rails, Bash, Maven, GIT, SVN, Bamboo/Jira/Confluence CLI, Nexus/Artifactory REST/XML, SDLC, CI/CD, Gerrit/Subversion, MySQL/PostgreSQL, Crowd/OpenID, Capistrano

8.2 Configuration management and deployment automation with Puppet and Capistrano

8.3 Continuous Integration (CI) and Continuous Deployment (CD) with Bamboo and Jenkins

8.4 Performance Monitoring System for Jetty, JVM, Linux and Splunk

8.5 Secured FTP server with virtual users and admins users with complex access permissions

8.6 BOM (Bill Of Materials) Administration and Management Application (BAMA) Enhancement and maintenance

8.7 Two Steps bare-metal Linux bootstrap system for HP Servers

8.8 Linux HA for Git Repo(Gerrit) and Maven Repo (Artifactory/Nexus) Servers

8.9 Automation scripts and tools for DevOps

8.10 CoreOS and Docker prototyping and testing

8.11 SDLC process management and junior team member training and mentoring

9. Nov. 2006 - Oct. 2010, Senior Software Engineer, Boomi Inc., Berwyn, PA, USA

As the market and technology leader in on-demand integration, Boomi is the industry's first and leading integration platform-as-a-service and a two-time SIIA CODiE award winner - Best Application Integration Solution (2010) and Best On-Demand Platform (2009).

9.1 Boomi On-Demand Business Integration Cloud Service

9.2 Cloud Platform/Infrastructure Design and Implementation for Boomi On Demand Production Environment